DTI5126 Assignment 1:

Overview

This assignment is worth 100 points and is solved in **groups of 2 students**. Both members receive the same mark. It covers Modules 0 to 3: data types, data quality, cleaning, data partitioning and OLAP.

Learning objectives. By the end of this assignment, you should be able to:

1. Compute and interpret basic descriptive statistics by hand.
2. Load, inspect, clean and transform a real dataset in R using the functions shown in class.
3. Partition data into training, validation and test sets.
4. Build a small OLAP cube in R and apply roll-up, slice and dice.
5. Read a research paper on data cleaning and discuss it critically, with proper references.

R level. Use base R as shown in the lectures and `Dataprep.R` (`read.csv`, `summary`, `sapply`, `table`, `aggregate`, `model.matrix`, `apply`/`tapply`, `hist`, `boxplot`, `plot`, `pairs`). The dplyr verbs from Tutorial 1 (`select`, `filter`, `mutate`, `group_by`, `summarize`) are also allowed. Other packages are optional and must be named in your report.

Deliverables (upload all three to Brightspace):

File	Contents
`Lastname1_Lastname2_A1.pdf` (or .docx)	Answers to all parts, with R output and plots pasted in, and a cover page with both names and student numbers
`Lastname1_Lastname2_A1.R`	One commented R script that runs from top to bottom without errors and reproduces every result in the report
`Lastname1_Lastname2_A1.csv`	The dataset used in Part 2 (or a link to it in the report if it is larger than 20 MB)

Rules.

- Use `set.seed()` with a value of your choice before any random step, so results can be reproduced.
- Label every question with its number (e.g. Q2.7) in both the report and the script.
- AI tools cannot be used to solve this assignment. If detected, the submission gets a zero and is reported to the academic unit (see the course outline).
- Questions about the assignment go to the TA first.

Part 1: Concepts and calculations (20 points)

Answer in the report. Show your working for calculations; you may check numeric answers in R (Q1.3 and Q1.4), but the working must be shown. Questions are adapted from Han, Kamber and Pei, *Data Mining: Concepts and Techniques*, 3rd ed., Chapters 2 and 3.

Q1.1 Data quality (4 points) *(Han Ex. 3.1)*. Name four dimensions of data quality covered in class (e.g. accuracy, completeness). For two of them, give an example where the same dataset is good enough for one use but not for another.

Q1.2 Missing values (4 points) *(Han Ex. 3.2)*. Describe four methods for handling tuples with missing values. For each, give one situation where it is a poor choice (hint: the fraud detection example from Lecture 2).

Q1.3 and Q1.4 use the following age values (in increasing order):

```
13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35,
36, 40, 45, 46, 52, 70
```

Q1.3 Descriptive statistics (6 points) *(Han Ex. 2.2)*.

1. Compute the mean, median and midrange. (1.5)
2. Give the mode(s) and state the data's modality. (1)
3. Find Q1 and Q3 and give the five-number summary. (1.5)
4. Draw a boxplot (by hand or with `boxplot()`) and say, using the 1.5 × IQR rule, whether any value is an outlier. (2) (you will need to do some research to understand with is Q1, Q2 and IQR)

Q1.4 Transformation (6 points) *(Han Ex. 3.3, 3.7)*.

1. Smooth the data by bin means using equal-frequency bins of depth 3. Show the bins. (2)
2. Partition the data into 3 equal-width bins. Give the bin boundaries and counts, and say which binning method better represents this data and why. (2)
3. Normalize age = 35 using min-max normalization to [0, 1], and using z-score normalization (use the sample standard deviation). (2)

Part 2: Preparing your own dataset in R (55 points)

Pick a public dataset (e.g. Kaggle, UCI Machine Learning Repository, Statistics Canada, Ottawa Open Data) or one from your own research. It must have **at least 500 rows and 8 columns, including at least 4 numeric and 2 categorical columns**. Datasets used in class (WestRoxbury, BostonHousing, titanic, mtcars, iris) are not allowed. For every question, paste the R output (or plot) in the report and explain it in your own words.

Q2.1 Describe the dataset (4 points). What does it measure, who collected it and how, and where you got it (link). State the unit of observation (what one row is), the file format and delimiter, and one research question you would like this data to answer.

Q2.2 Load and inspect (4 points). Read the file into a data frame (e.g. `read.csv`) and explain any options you used. Show `dim`, `head`, `str` and the class of each column. Rename at least two columns to clearer names, and convert categorical columns to factors (ordered where it makes sense). Do not remove missing values yet.

Q2.3 Summary statistics (5 points). Using `sapply`, build one data frame with the mean, standard deviation, min, max, median and number of missing values for each numeric column (as in Lecture 2). Use `table()` on one categorical column, and `aggregate()` to give the mean of one numeric column per category. Comment on two things you learned.

Q2.4 Missing values (6 points).

1. Count the missing values in each column. If there are fewer than 2 columns with missing values, set about 5% of the values in two columns (one numeric, one categorical) to `NA` using `sample()`, and say so. (1)
2. Replace missing numeric values with the column median and missing categorical values with the most frequent category. Show the summary before and after. (3)
3. Give one case in your data where deleting rows (`na.omit` or `complete.cases`) would be a better choice than imputing, or explain why it would not be. (2)

Q2.5 Outliers (5 points). For two numeric columns, find outliers with the 1.5 × IQR rule and with |z-score| > 3. Report how many each method flags, show a boxplot, and decide whether to keep, correct or remove them for your research question.

Q2.6 Transformation (8 points).

1. Write your own R function `minmax <- function(x) {...}` and use it, and `scale()`, to normalize one numeric column. Check the range of each result. (3)
2. Discretize a different numeric column into equal-width bins with `cut()` or `.bincode()`, and show the count per bin with `table()`. Explain your choice of number of bins. (2)
3. Convert one categorical column into dummy variables with `model.matrix()`, as in class. Drop one dummy and explain why. (3)

Q2.7 Construction and reduction (4 points). Create one new attribute from existing columns (e.g. a ratio or an age from a date) and explain what it adds. Compute the correlation matrix of the numeric columns with `cor()` and use it, plus your own judgement, to identify one redundant or irrelevant column to drop.

Q2.8 Visualization (8 points). Make four plots in base R (or ggplot2 if you prefer): a histogram, a boxplot of a numeric column by category, a scatterplot of two numeric columns, and a scatterplot matrix with `pairs()` of at least four numeric columns. Each plot needs a title and clear axis labels with units. For each, write two or three sentences on what it shows about your research question. (2 per plot)

Q2.9 Data partitioning (4 points). After `set.seed()`, partition the data into training (60%) and validation (40%), then separately into training (50%), validation (30%) and test (20%), using `sample()` and `setdiff()` as in `Dataprep.R`. Report the number of rows in each set, compare the mean of one numeric column across the sets, and explain why we need a separate test set.

Q2.10 OLAP cube (7 points). Using Tutorial 2 as a guide, choose a numeric measure and three categorical dimensions from your data (you may use your bins from Q2.6 or a year/month from a date as a dimension).

1. Build the cube with `tapply()` and show `dimnames()`. (2)
2. Show one roll-up, one slice and one dice with `apply()` and indexing. (3)
3. In one sentence each, say what business or research question each operation answers. (2)

If your dataset does not have three usable dimensions, use the `sales_fact` table generated in Tutorial 2 for this question and say so.

Part 3: Research on data cleaning (15 points)

Read: X. Chu, I. F. Ilyas, S. Krishnan and J. Wang, "Data cleaning: Overview and emerging challenges," in *Proc. 2016 Int. Conf. on Management of Data (SIGMOD)*, 2016, pp. 2201–2206.

Write in your own words; quotations must be marked and cited. Use IEEE reference style.

Q3.1 Detection and repair (5 points). In two paragraphs, explain the main error detection approaches the paper describes (qualitative versus quantitative) and the main error repair approaches. Give one concrete example for each.

Q3.2 Link to your data (4 points). Which types of errors from the paper did you find, or could you expect, in your Part 2 dataset? Which detection or repair approach from the paper would you use for them, and how does it compare with what you did in Q2.4 and Q2.5?

Q3.3 Data cleaning for machine learning (6 points). In two paragraphs, discuss why data cleaning matters for machine learning models (e.g. effect on accuracy, bias, or training cost). Support your discussion with **at least two peer-reviewed references published in 2022 or later**, other than the Chu et al. paper. Websites and blogs do not count.

Marking rubric (100 points)

Part	Questions	Points
Part 1: Concepts and calculations	Q1.1 (4), Q1.2 (4), Q1.3 (6), Q1.4 (6)	20
Part 2: R on your own dataset	Q2.1 (4), Q2.2 (4), Q2.3 (5), Q2.4 (6), Q2.5 (5), Q2.6 (8), Q2.7 (4), Q2.8 (8), Q2.9 (4), Q2.10 (7)	55
Part 3: Research	Q3.1 (5), Q3.2 (4), Q3.3 (6)	15
Code and report quality	Script runs top to bottom (4), code is commented and labelled by question (2), report is organized and plots are readable (2), references in IEEE style (2)	10
Total		**100**

Each question is marked on three levels:

Level	Share of points	What it looks like
Complete	100%	Correct result, R output shown, and a clear explanation in your own words tied to your data
Partial	50–70%	Correct code or result but missing or weak explanation, or a small error
Minimal	0–30%	Missing output, wrong method, or explanation that does not match the result

Deductions: −5 if file names do not follow the required format, and the late policy announced on Brightspace.

Submission checklist

- ☐ Cover page with both names and student numbers
- ☐ Every question labelled (Q1.1 to Q3.3) in the report and the script
- ☐ `set.seed()` used before every random step
- ☐ Script runs from a clean R session with only the CSV in the working directory
- ☐ Dataset has at least 500 rows, 8 columns, 4 numeric and 2 categorical
- ☐ At least two peer-reviewed references from 2022 or later in Part 3
- ☐ Three files uploaded: report, `.R` script and `.csv`